Python Programming

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Python Programming (A35321)

Short Title:	Python Programming
Faculty	Science and Computing
Department	Computing and Mathematics
Cluster	Software and Web Development
Author	ADAVY
Credits	5

Level: Introductory

Description of Module / Aims

This course focuses on the basic knowledge of Python and small project examples, which aims to help students who needs to learn data science and big data technology quickly master basic Python programming knowledge, be able to write properly running programs and develop projects of interest. After learning this course, students should be able to master basic Python programming concepts, develop good programming habits, and continue to learn advanced Python technologies.

Programmes

`COMP-0678`	BSc (Hons) in Applied Computing (International) (`WD_KACMI_B`)	2 / 3 / M

Indicative Content

- Programming Environment Setup: to set up the environment in Linux, OS X and Windows Variables and Simple Data Types: variables, character strings, figures, annotations
- Introduction to Lists: lists, modifications, additions, deletion of elements, organization of lists
- Operating Lists: traversing lists, creating lists of figures, slicing lists and copying lists, tuples
- IF Statements: conditional statements that uses IF statements to process lists
- Dictionaries: creating dictionaries, as well as modifying, deleting, traversing and nesting dictionaries
- Loops: while loops, and using loops to process lists and dictionaries Functions: definitions, parameters, returned values, modules
- Classes: creating classes, inheriting classes, importing classes, Python Standard library Files and Exceptions: file reading and writing, exception handling, data storage
- Test Functions: test functions, test classes
- Project 1 Creating Games: project planning, package installation, creation of instances and methods
- Project 2 Data Visualization: package installation, data download, drawing of various diagrams, etc.
- Project 3 Small Web Applications: Web framework usage, functionality addition and project deployment.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. 1. Master the basic knowledge of script language programming, master the basic method of programming, master the basic theory, method and application of programming, master the relevant basic provisions of national standards for advanced programming.
2. Be able to use Python to solve practical application problems. Cultivate students' computational thinking ability, innovation ability and ability to find, analyze and solve problems.
3. Be able to use Python correctly and skillfully for program design; Be able to read and write programs with more complexity;.

Learning and Teaching Methods

- This module will be presented by a combination of lectures and computer-based practicals whilst capitalising on a web-enhanced learning environment.
- The lectures will be used to introduce new topics and their related concepts.
- A cooperative learning/peer tutoring (i.e. pair-programming, problem-based learning for one assignment) approach will be adopted during the practical sessions.
- Self-directed learning will be encouraged throughout the duration of the module.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3
Assignment	30%	1,2,3
In-Class Assessment	10%	1,2,3
Project	60%	1,2,3

Assessment Criteria

<40%: Inability to design, develop and test maintainable, persistent, robust UI applications to solve a particular problem

40%-49%: Ability to design, develop and test maintainable, persistent, robust UI applications to solve a straight-forward problem.

50%-59%: Comfortable with designing, developing and testing maintainable, persistent, robust UI applications to solve problems similar to those presented in the module.

60%-69%: Proficient with designing, developing and testing maintainable, persistent, robust, high-quality UI applications to solve complex problems.

70%-100%: Proficient with designing, developing and testing maintainable, persistent, robust, high-quality, elegant UI applications to solve complex problems that are substantially different to those studied in the module.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	82	

Essential Material

- Halterman, R. _Fundamentals of Python Programming_. UK: Southern Adventist University (2019), and Internet Archive, 2019.

Supplementary Material

- "W3 Schools." W3 Schools on-line Web Tutorials. https://www.w3schools.com/python/